

Project Overview

AI-Powered Document ChatBot

The project aims to develop an AI-powered Document ChatBot that enables users to upload PDF and DOCX files and interact with them using natural language. Instead of manually searching through documents, users can ask questions, and the system retrieves the most relevant information before generating accurate responses using a Retrieval-Augmented Generation (RAG) pipeline.

The application consists of:

- **Backend:** Processes uploaded documents, extracts text, generates embeddings, performs semantic search, and generates AI responses.
- **Frontend (Planned):** A user-friendly web interface where users can upload documents, manage files, and chat with the AI through an interactive chat window.

2. Technology Stack

Category	Technologies
Frontend	React.js, Vite, HTML5, CSS3, JavaScript
Backend	Python, FastAPI, REST APIs, Swagger/OpenAPI
AI & NLP	Groq API (Llama 3.3 70B), Sentence Transformers (all-MiniLM-L6-v2), Retrieval-Augmented Generation (RAG)
Document Processing	PyMuPDF (PDF), python-docx (DOCX), OCR (Tesseract OCR/PaddleOCR for scanned PDFs – Planned)
Vector Database	ChromaDB
Relational Database	PostgreSQL (Production) / SQLite (Development)
Authentication & Security	JWT Authentication, OAuth2, Password Hashing (bcrypt), Role-Based Access Control (RBAC)
Storage	Local File Storage (Development), Cloud Storage (AWS S3/Azure Blob Storage – Planned)
Development Tools	Git, GitHub, Visual Studio Code (VS Code), python-dotenv
Testing & API Documentation	Swagger UI (OpenAPI), Postman
Deployment	Docker, Nginx, Render/Railway/Azure (Backend), Vercel/Netlify (Frontend)

3. Project Roadmap

Phase 1 – Project Setup [DONE]

- Project structure
- Environment setup
- Dependency management

Phase 2 – Document Processing [DONE]

- PDF parsing
- DOCX support
- Text cleaning & chunking

Phase 3 – AI Pipeline [DONE]

- Embedding generation
- ChromaDB integration
- Semantic search
- Groq LLM integration

Phase 4 – API Development [CURRENTLY WORKING ON THIS]

- Upload API
- Chat API
- Duplicate document detection
- Error handling
- Document management APIs

Phase 5 – Frontend Development [TO-DO]

- User dashboard
- File upload interface
- Chat interface
- Document management

Phase 6 – Advanced Features [TO-DO]

- Multi-user support
- OCR for scanned PDFs
- Chat history
- Authentication
- Deployment & testing